

The impact of climate change on the hydrology of the Wien river and the implications for flooding of the adjacent U4 subway line

TEACHBOOKS

March 7, 2025

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0.1 BEP Thirza

0.1.1 Introduction

Last September, the Wien River, normally a small trickle, swelled into a raging torrent, having a serious impact on public transport. Damage to the public transport was kept to a minimum thanks to the flood alarm plan, but still almost one-million-liter water had to be pumped out of the subway lines. (Krone.At, 2024b) The Wien River is one of the largest rivers of Vienna and has a catchment area of 230 km². Along a reach of over 8 km, the subway line U4 follows the river in an open section on the right bank. Flooding on the Wien River is critical because of the large area of impervious surfaces in the catchment area. Wien River floodings are typically flash floodings. The threat regarding flooding in Vienna is caused by large channel slopes and flow velocities, rapid increase of discharge and the absence of natural retention areas. Furthermore, the low hills and mountains in the area intensify storm events compared to plain areas through intensified convective air movement. (Compton et al., 2009, pp. 13–14) Dore (2005) states in his article about climate change and changes in global precipitation patterns that precipitation is expected to increase in the Northern Hemisphere. The wet areas will get wetter, and the rainfall will get more intense. Changes in precipitation due to climate change may have influence on the water level and discharge in the Wien River. This raises concern about the river in relation to the adjacent U4 subway line.

Flood risk management is critical for urban resilience, especially in places where natural waters are closely linked to infrastructure. Compton et al. (2009) highlight in their study on uncertainty and disaster risk management how an approach based on catastrophe modeling can provide a useful framework for comparing different mitigation strategies as well as integrate the risk perspectives of different technical disciplines. For their study they use hydraulic models to obtain probability of failure for different storm return periods and states of the flood control reservoirs. This report will focus on flood risk as well but shifts the focus towards how the discharge and water levels in the Wien River will change due to climate change, influencing the flood risk for the U4 subway line. Hydrological models with different climate change scenarios will be implemented to simulate the future discharge scenarios of the Wien River. The report will focus solely on changes in flood frequency due to climate change and excludes the current flood protection infrastructures and adaption measures these might require. This research will contribute to Vienna's flood risk management and is relevant for the transportation authorities, policymakers and urban developers in Vienna who are seeking to make the city future-proof for increasing threats due to climate change.

In this report the following research question will be answered: “What is the impact of climate change on the hydrology of the Wien River and what are the implications for flooding of the adjacent U4 subway line?” This is done using the following sub-questions:

- What is the current maximum discharge (m³/s) in the Wien above which the U4 subway line will flood?
- How often does the discharge currently exceed the maximum?
- How often will the discharge exceed this maximum in the future under different climate change scenarios?

The research question will be answered using the sub-questions. To determine the maximum discharge threshold for flooding of the U4, literature study needs to be done. Historical events, flood risk maps, research papers and monitoring stations need to be analyzed. The current frequency at which the discharge exceeds this maximum can be determined using eWaterCycle. Past streamflow can be simulated with a hydrological model. The simulated discharge data then can be compared to the flood threshold determined before. The model should be calibrated using real-world data. The frequency of flooding of the U4 with current discharge data can be conducted with return period calculations. The future discharge can be simulated with the use of future climate projections. Different IPCC scenarios give different projected climate variables which can be used in hydrological models using eWaterCycle to simulate future discharge. These modeled future discharge scenarios should be compared to the flood threshold to estimate the future frequency of exceedance for different climate change scenarios.

In Chapter 2 the current maximum discharge (m³/s) is determined above which the U4 subway line will flood. Chapter 3 determines how often the discharge currently exceeds this maximum using hydrological models. In Chapter 4 different climate change scenarios are projected in hydrological models to simulate future discharge. These future discharge scenarios are compared to the flood

threshold to estimate the future flooding frequency of the U4 subway line. In Chapter 4 the conclusion and discussion can be found.

0.1.2 Chapter 1: Current maximum discharge before flooding of the U4

The Wien River finds its origin in the Wienerwald, west of Vienna, and enters the city after approximately 20 km. The river discharges into the Donaukanal. The historical hydrology of the Wien River cannot be reconstructed with certainty. However, before the construction of intercepting sewers along the river in the 1830s and its regulation and channelization at the end of the 19th century, the estimated mean annual discharge was approximately 2 m³/s. (Pollack et al., 2016)

Since the flood retention basins that were made in the early 1800's, the 10-year return flood was estimated at 140 m³/s, and the 100-year return flood at 200 m³/s. Due to the high potential losses in the city of Vienna, the Wien River is designed to withstand a 1000-year discharge return period (Faber & Nachtnebel, 2002).

Compton et al. (2009, p. 54) state: "A failure that results in the release of water to the U4 occurs when the discharge into the Wien River exceeds the given threshold, resulting either in overtopping of the floodwall or collapse of the floodwall due to either foundation scouring or hydrostatic pressure." While failure due to overtopping is a function of the flow rate in the channel, uncertainties in the water flow rate are expected to be minimal since the Wien River is a channelized river with well characterized geometry. More uncertainty is expected in the erosive failure and wall collapse which are a function of the shear at the channel bed and the shear strength of the invert.

According to the report, the failure leading to overflowing of the U4 is expected to occur at a discharge of 530 m³/s. Given the uncertainties in the floodwall's resistance parameters, this critical discharge is modeled as a normal distribution with mean value of 530 m³/s and a standard deviation of 10 m³/s. This means that at a discharge of around 510 m³/s failure could occur with a probability of 5 percent, and at a discharge of around 550 m³/s with a probability of 95 percent. (Compton et al., 2009)

Faber (2006) also analyzed flood risk in an Austrian context, and specifically for the Wien River. He estimated peak flow frequencies using a rainfall-runoff model and Monte Carlo simulations to account for uncertainties. A total of 7000 simulations were performed within the critical range of 400 to 600 m³/s where failures were most likely to occur. Figure 1 shows that the flood walls can handle discharges up to 500 m³/s, while overtopping of the floodwall is almost certain at a discharge of 560 m³/s. The mean value of the discharge capacity before overtopping amounts to 534 m³/s, with a standard deviation of 14 m³/s. The mean value of discharge for structural floodwall failure is 541 m³/s with a standard deviation of 16 m³/s (Faber, 2006).

Faber also analyzed the probabilities of the different failure events, overtopping and structural damage of the flood wall, individually and in combination. The probability of structural flood wall failure without overtopping was not observed in any simulation. The overall system reliability is above 99 percent, which indicates that failure of the flood wall is extremely rare with the used past peak flows. He further analyzed the return periods of the failure events. The installation of the controlled retention basins in 1998 increased the return period of failure from approximately 550 years to 1100 years. This return period exceeds the 1000-year discharge return period the Wien River is designed for (Faber & Nachtnebel, 2002). This deviation is due to limitations of the return period-based design, which does not fully account for uncertainties in flood frequency and magnitude.

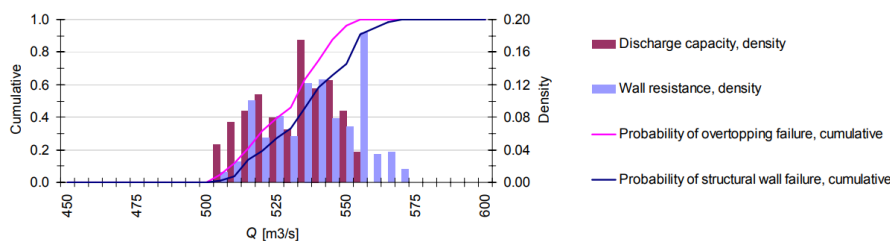


Figure 1: "Distribution of the resistance of the hydraulic system in terms of the bankfull discharge and the flow related to structural floodwall failure" (Faber, 2006)

It can be concluded that the critical event for flooding of the U4 subway line is overtopping rather than structure failure of the flood wall. According to Faber (2006), the mean failure discharge at which overtopping happens is 534 m³/s, with a 5 percent probability at 511 m³/s, and a 95 percent probability at 557 m³/s. These values align with the threshold values determined by Compton et al. (2009), who estimated a mean failure discharge of 530 m³/s, with 5 and 95 percent probabilities of failure

at discharges of approximately 510 and 550 m³/s. The threshold values are normally distributed, so looking at 1 critical threshold value would be a simplification. The probability of exceeding a threshold is heavily dependent on where that threshold is situated in the normal distribution. For this reason the return periods of all discharges in the normal distribution are calculated. This will give an indication of the return period of a certain discharge, and the probability that this specific discharge will lead to flooding of the U4 subway line.

0.1.3 Chapter 2: Current frequency of threshold exceedance

In this chapter the current frequency of exceeding the threshold, determined in chapter 2, will be analysed. This is done by looking at the available observation data of the catchment area of the Wien River. This observation data is available through eWaterCycle. As determined in chapter 2, the Wien River is designed for a 1000-year discharge return period. The observation data is unlikely to cover a period of 1000 years, so we will need to extrapolate it to estimate the discharge corresponding to a 1000-year return period and determine the return period of the previously established threshold.

General First of all, some general python and eWaterCycle libraries need to be imported:

```
# general python
import warnings
warnings.filterwarnings("ignore", category=UserWarning)

import numpy as np
from pathlib import Path
import pandas as pd
import matplotlib.pyplot as plt
import xarray as xr

#niceties
from rich import print
import seaborn as sns
sns.set()

# general eWaterCycle
import ewatercycle
import ewatercycle.models
import ewatercycle.forcing
```

eWaterCycle provides access to the Caravan dataset. This dataset contains data on rainfall, potential evaporation and discharge for all the catchments in the different Camel datasets. The Caravan dataset contains a Camel dataset of the catchment of the Wien River. This catchment area is loaded below:

```
camelsgb_id = "lamah_208082"
```

The start and end date of the experiment have to be specified. The start and end date of the available observation data are determined further below, and are hardcoded in the following cell:

```
experiment_start_date = "1981 -08 -01T00:00:00Z"
experiment_end_date = "2020 -12 -31T00:00:00Z"
```

The forcing data can be generated or previously generated data can be loaded:

```
# Even though we don't use caravan data as forcing, we still call it forcing
# because it is generated using the forcing module of eWaterCycle
forcing_path_caravan = Path.home() / "forcing" / camelsgb_id / "caravan"
forcing_path_caravan.mkdir(exist_ok=True, parents=True)

# If someone has prepared forcing for you, this path needs to be changed to that location.
prepared_forcing_path_caravan_central = Path("location/of/forcing/data")
```



```

# option one: generate forcing data
# camelsgb_forcing = ewatercycle.forcing.sources['CaravanForcing'].generate(
#     start_time=experiment_start_date,
#     end_time=experiment_end_date,
#     directory=forcing_path_caravan,
#     basin_id=camelsgb_id,
# )

# option two or three: load data that you or someone else generated previously
camelsgb_forcing = ewatercycle.forcing.sources['CaravanForcing'].load(Path("/home/thirza/forci

print(camelsgb_forcing)

CaravanForcing(
    start_time='1981 -08 -01T00:00:00Z',
    end_time='2025 -08 -31T00:00:00Z',
    directory=PosixPath('/home/thirza/forcing/lamah_208082/caravan'),
    shape=PosixPath('/home/thirza/forcing/lamah_208082/caravan/lamah_208082.shp'),
    filenames={
        'tasmax': 'lamah_208082_1981 -08 -01_2025 -08 -31_tasmax.nc',
        'tas': 'lamah_208082_1981 -08 -01_2025 -08 -31_tas.nc',
        'tasmin': 'lamah_208082_1981 -08 -01_2025 -08 -31_tasmin.nc',
        'Q': 'lamah_208082_1981 -08 -01_2025 -08 -31_Q.nc',
        'pr': 'lamah_208082_1981 -08 -01_2025 -08 -31_pr.nc',
        'evspsblpot': 'lamah_208082_1981 -08 -01_2025 -08 -31_evspsblpot.nc'
    }
)

```

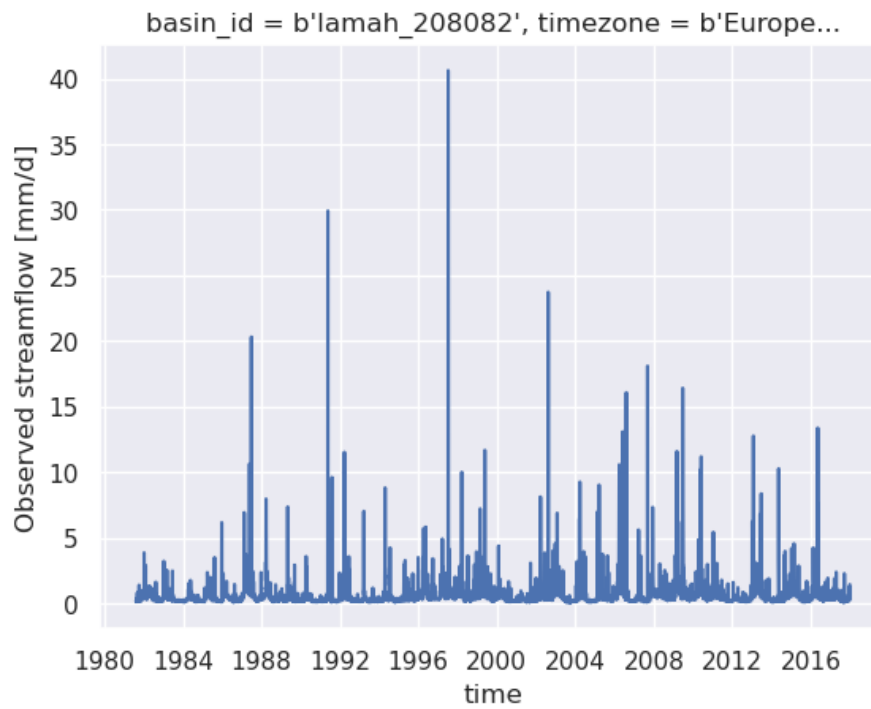
Above, it can be seen that the forcing data contains precipitation, potential evaporation, discharge and the near-surface temperatures (tas). For this research, only the discharge data is relevant. The discharge data is loaded from the forcing below. The data contains the maximum discharge values per day.

```

#quick plot of the discharge data.
ds_forcing = xr.open_mfdataset([camelsgb_forcing['Q'], ])
ds_forcing["Q"].plot()
print(ds_forcing['time'].values)

['1981 -08 -01T00:00:00.000000000' '1981 -08 -02T00:00:00.000000000'
 '1981 -08 -03T00:00:00.000000000' ... '2020 -12 -29T00:00:00.000000000'
 '2020 -12 -30T00:00:00.000000000' '2020 -12 -31T00:00:00.000000000']

```



Since the threshold values determined in chapter 2 are in m³/s, the observed discharge data, now in mm/d, is converted to m³/s as well.

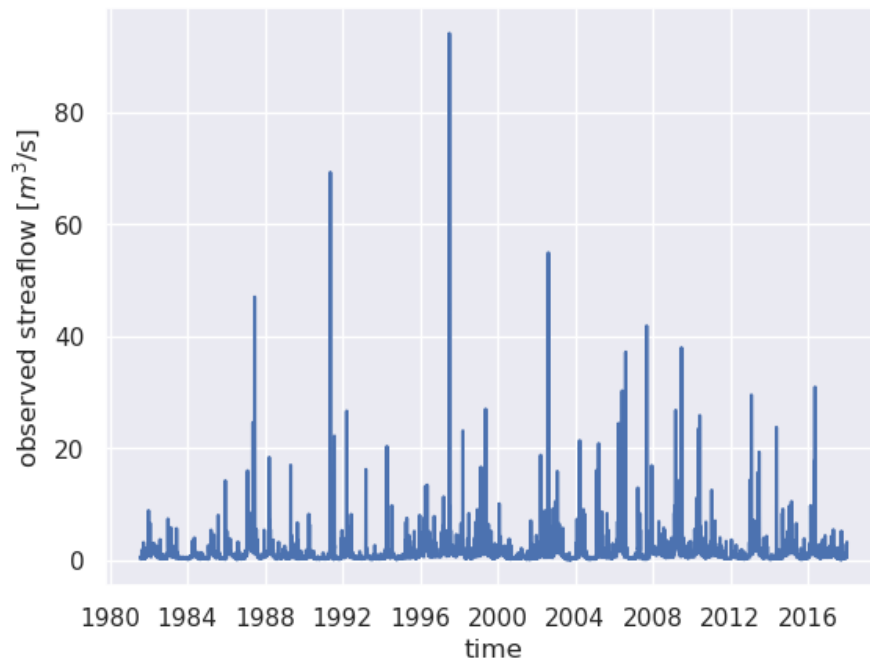
```
catchment_area = ds_forcing["area"].values

# go from mm/day to m3/s
discharge = []
for i in range(len(ds_forcing["Q"].values)):
    discharge.append((ds_forcing["Q"].values[i] * catchment_area * 1000) / (24 * 60 * 60))

x = ds_forcing["time"].values
y = discharge

plt.plot(x,y)
plt.xlabel('time')
plt.ylabel('observed streaflow [m3/s]')

Text(0, 0.5, 'observed streaflow [m3/s]')
```



The maximum discharge of the observed data is:

```
print(f'The maximum observed discharge in the observed data is {max(discharge):.3f} m3/s')
```

The maximum observed discharge in the observed data is 94.052 m3/s

This discharge can be used to calculate the return periods of the normally distributed threshold values. This is done using the Generalized Extreme Value distribution. The return period of the mean threshold value of 530 m3/s is calculated below:

nog meer uitleggen en aanpassen!!!

```
# Find the maximum annual discharge
max_discharge = ds_forcing["Q"].groupby("time.year").max()
maxdischarge = max_discharge.values * catchment_area * 1000 / (24 * 60 * 60) #convert to m3/s
# print(maxdischarge)

import scipy.stats as stats
import math
# # discharge data > 20 m3/s
# data = [value for value in discharge if value > 20]

# all discharge data
cleaned_discharge = [x for x in discharge if not math.isnan(x)]
data = cleaned_discharge

# # maximum annual discharge
# data = [x for x in maxdischarge if not math.isnan(x)]
# # print(data)

# Generalized Extreme Value (GEV) distribution
shape, loc, scale = stats.genextreme.fit(data)

# Define threshold value
threshold = 530
```

```

# calculate exceedance probability in years
p = 1 - stats.genextreme.cdf(threshold, shape, loc=loc, scale=scale)
# calculate return period in years
T = 1 / (p * 365.25)
# print(T)

# calculate change threshold exceedance at least once in 1000 and 1100 years
P_1000 = 1 - (1 - p) ** 1000
P_1100 = 1 - (1 - p) ** 1100

# Results 1000
print(f"Returnperiod for 530 m³/s: {T:.2f} years")
print(f"Chance of at least one exceedance in 1000 years: {P_1000:.5f} ({P_1000 * 100:.2f}%)")
print(f"Chance of at least one exceedance in 1100 years: {P_1100:.5f} ({P_1100 * 100:.2f}%)")

# expected amount of times threshold is exceeded in 1000 years
expected_exceedances1000 = 1000 / T if T > 0 else 0
print(f"The discharge currently exceeds the threshold of 530 m³/s {expected_exceedances1000:.3f} times")

# expected amount threshold is exceeded in 1100 years
expected_exceedances1100 = 1100 / T if T > 0 else 0
print(f"The discharge currently exceeds the threshold of 530 m³/s {expected_exceedances1100:.3f} times")

Returnperiod for 530 m³/s: 168.64 years

Chance of at least one exceedance in 1000 years: 0.01610 (1.61%)

Chance of at least one exceedance in 1100 years: 0.01770 (1.77%)

The discharge currently exceeds the threshold of 530 m³/s 5.930 times in 1000 years

The discharge currently exceeds the threshold of 530 m³/s 6.523 times in 1100 years

import scipy.stats as stats
import math
# # discharge data > 20 m³/s
# data = [value for value in discharge if value > 20]

# # all discharge data
# cleaned_discharge = [x for x in discharge if not math.isnan(x)]
# data = cleaned_discharge

# maximum annual discharge
data = [x for x in maxdischarge if not math.isnan(x)]
# print(data)

# Generalized Extreme Value (GEV) distribution
shape, loc, scale = stats.genextreme.fit(data)

# Define threshold value
discharge = np.linspace(1, 600, 600)

return_period = []
prob = []
for i in range(len(discharge)):

```

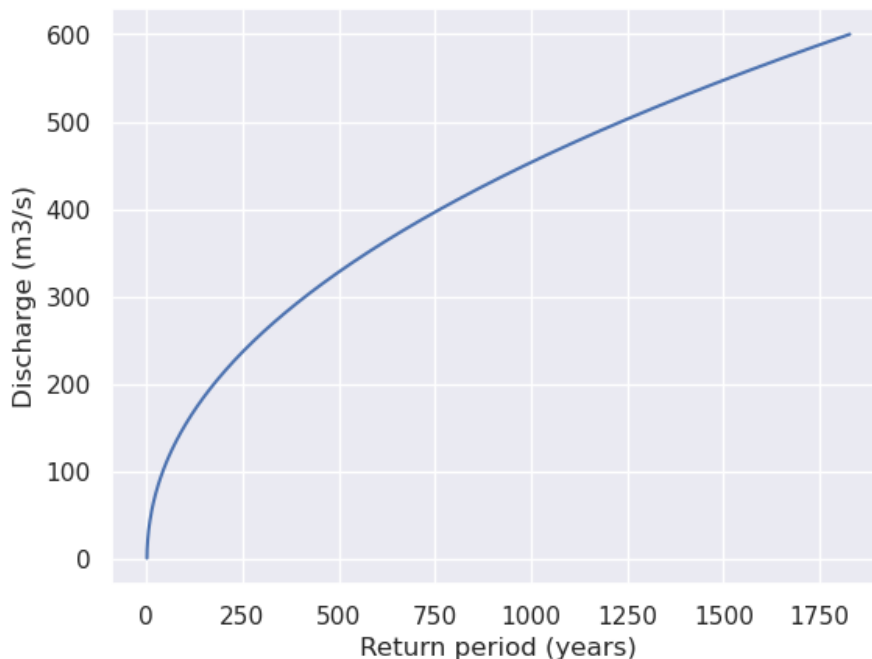
```

# calculate exceedance probability in years
p = 1 - stats.genextreme.cdf(discharge[i], shape, loc=loc, scale=scale)
prob.append(p)
# calculate return period in years
T = 1 / (p )
return_period.append(T)
# print(return_period)

plt.plot(return_period, discharge)
plt.xlabel('Return period (years)')
plt.ylabel('Discharge (m3/s)')

Text(0, 0.5, 'Discharge (m3/s)')

```



Below both the normal distribution of the exceedance threshold values and the return periods of discharges are plotted below. This graph shows the return period and the probability density are related to the discharge. A discharge value more to the right of the normal distribution, has a higher return period, but also has a higher probability of actually causing flooding of the U4 subway line.

```

#1
mean = 530
p5 = 510 # 5th percentile
p95 = 550 # 95th percentile

#Calculate standard deviation
z_5 = stats.norm.ppf(0.05) # -1.645
z_95 = stats.norm.ppf(0.95) # 1.645
std_dev = (p95 - mean) / z_95 # \sigma = (550 - 530) / 1.645
x = np.linspace(0, 600, 1000)
y = stats.norm.pdf(x, mean, std_dev)

#2
mean_2 = 534
p5_2 = 511 # 5th percentile
p95_2 = 557 # 95th percentile

```

```

#Calculate standard deviation
z_5 = stats.norm.ppf(0.05) # -1.645
z_95 = stats.norm.ppf(0.95) # 1.645
std_dev_2 = (p95_2 - mean_2) / z_95 # \sigma = (550 - 530) / 1.645
x_2 = np.linspace(0, 600, 1000)
y_2 = stats.norm.pdf(x_2, mean_2, std_dev_2)

fig, ax1 = plt.subplots(figsize=(15, 5))

#plot
# ax1=plt.figure(figsize=(8, 5))
ax1.plot(x, y, label="mean = 530", color='blue')
ax1.plot(x_2, y_2, label="mean = 534", color='red')

# # Mark the 5th and 95th percentiles
# plt.axvline(p5, color='red', linestyle='dashed')
# plt.axvline(p95, color='red', linestyle='dashed')
# plt.axvline(mean, color='orange', linestyle='dashed')
# plt.axvline(p5_2, color='green', linestyle='dashed')
# plt.axvline(p95_2, color='green', linestyle='dashed')
# plt.axvline(mean_2, color='orange', linestyle='dashed')

# Labels en titel
ax1.set_xlabel("Discharge (m3/s)")
ax1.set_ylabel("Probability density")
# plt.title(f"Normal distribution")
# plt.grid()

# Maak een tweede y -as
ax2 = ax1.twinx()

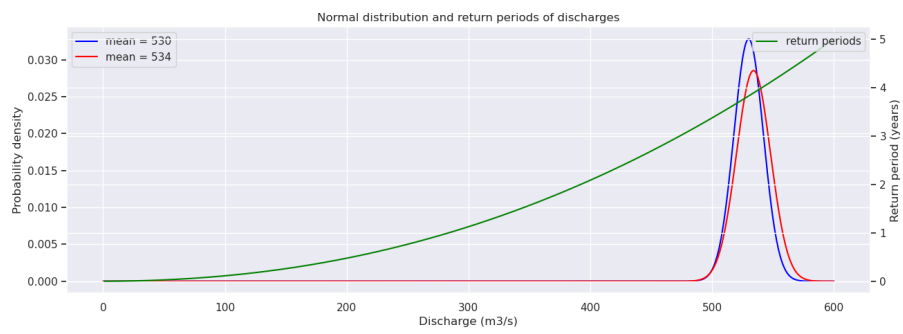
# Stel een andere functie voor de tweede y -as in (bijvoorbeeld lineaire functie)
y_linear = 0.1 * x # Voorbeeld lineaire functie

# Plot de lineaire functie op de tweede y -as
ax2.plot(threshold, return_period, label="return periods", color='green')
ax2.set_ylabel("Return period (years)")
ax2.tick_params(axis='y')

# Titel en legenda
plt.title(f"Normal distribution and return periods of discharges")
ax1.legend(loc="upper left")
ax2.legend(loc="upper right")

# Toon de plot
plt.grid(True)
plt.show()

```



0.1.4 Chapter 3: Calibration of the HBV model

In this chapter the future frequency of threshold exceedance is determined. This is done using different climate change scenarios, which are implemented in the HBV model. To be able to project the future discharge scenarios, the HBV model is used to model the future discharge. The HBV model needs to be calibrated using observation data. The model requires forcings, rainfall and potential evaporation, as inputs. With a set of parameters, which have to be calibrated, it can calculate modelled discharge at the outlet of the catchment.

The HBV model is available through eWaterCycle. The developer of the model, Sten Bergström (1992), says the HBV model can best be classified as a semi-distributed conceptual model. The model consists of three main components:

- subroutines for snow accumulation and melt
- subroutines for soil moisture accounting
- response and river routing subroutines

Precipitation and air temperature are the model inputs, and data on potential evapotranspiration is needed for the accounting of soil moisture. (Bergström, 1992)

General

First of all, some general python and eWaterCycle libraries need to be imported:

```
# general python
import warnings
warnings.filterwarnings("ignore", category=UserWarning)

import numpy as np
from pathlib import Path
import pandas as pd
import matplotlib.pyplot as plt
import xarray as xr

#niceties
from rich import print

#needed
from ipywidgets import IntProgress
from IPython.display import display

# general eWaterCycle
import ewatercycle
import ewatercycle.models
import ewatercycle.forcing
```

As stated in chapter 3, eWaterCycle provides access to the Caravan dataset, from which a Camel dataset of the catchment of the Wien River is loaded:

```
camelsgb_id = "lamah_208082"
```

The start and end date of the experiment have to be specified. The start and end date of the calibration have to be specified as well. The period of calibration is chosen to be around 75% of the experiment period. This means that the model is trained on 75% of the observation data, and can be tested on 25% of the observation data, to make sure the model is not only working for the data it was trained on, but on other periods of data as well.


```
# calibration dates op 75% van de experiment dates zetten!!!
```

```
experiment_start_date = "1981 -08 -01T00:00:00Z"
experiment_end_date = "2020 -12 -31T00:00:00Z"
```

```
calibration_start_time = experiment_start_date
calibration_end_time = "2010 -08 -31T00:00:00Z"
```

```
validation_start_time = "2010 -08 -31T00:00:00Z"
validation_end_time = experiment_end_date
```

The forcing data can be generated or previously generated data can be loaded:

```
# Even though we don't use caravan data as forcing, we still call it forcing
# because it is generated using the forcing module of eWaterCycle
forcing_path_caravan = Path.home() / "forcing" / camelsgb_id / "caravan"
forcing_path_caravan.mkdir(exist_ok=True, parents=True)
```

```
# If someone has prepared forcing for you, this path needs to be changed to that location.
prepared_forcing_path_caravan_central = Path("location/of/forcing/data")
```

```
# # option one: generate forcing data
# camelsgb_forcing = ewatercycle.forcing.sources['CaravanForcing'].generate(
#     start_time=experiment_start_date,
#     end_time=experiment_end_date,
#     directory=forcing_path_caravan,
#     basin_id=camelsgb_id,
# )
```

```
# option two or three: load data that you or someone else generated previously
camelsgb_forcing = ewatercycle.forcing.sources['CaravanForcing'].load(Path("/home/thirza/forci
```

```
print(camelsgb_forcing)
```

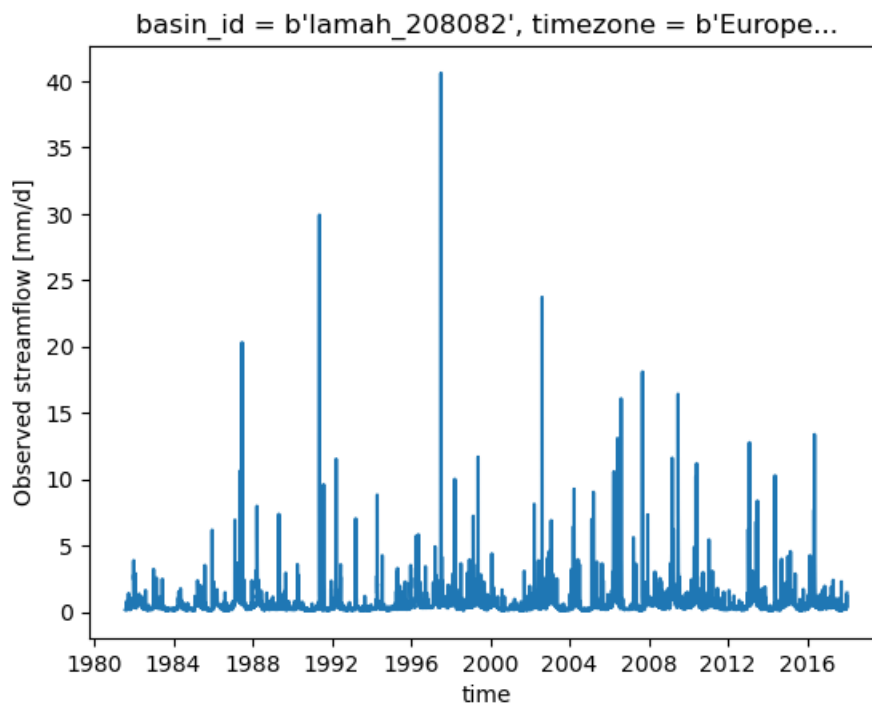
```
CaravanForcing(
    start_time='1981 -08 -01T00:00:00Z',
    end_time='2030 -12 -31T00:00:00Z',
    directory=PosixPath('/home/thirza/forcing/lamah_208082/caravan'),
    shape=PosixPath('/home/thirza/forcing/lamah_208082/caravan/lamah_208082.shp'),
    filenames={
        'tasmax': 'lamah_208082_1981 -08 -01_2030 -12 -31_tasmax.nc',
        'tas': 'lamah_208082_1981 -08 -01_2030 -12 -31_tas.nc',
        'tasmin': 'lamah_208082_1981 -08 -01_2030 -12 -31_tasmin.nc',
        'Q': 'lamah_208082_1981 -08 -01_2030 -12 -31_Q.nc',
        'pr': 'lamah_208082_1981 -08 -01_2030 -12 -31_pr.nc',
        'evspsblpot': 'lamah_208082_1981 -08 -01_2030 -12 -31_evspsblpot.nc'
    }
)
```

Above, it can be seen that the forcing data contains precipitation, potential evaporation, discharge and the near-surface temperatures (tas). For this research, only the discharge data is relevant. The discharge data is loaded from the forcing below.

```
#quick plot of the discharge data.
```

```
ds_forcing = xr.open_mfdataset([camelsgb_forcing['Q']])
ds_forcing["Q"].plot()
```

```
[<matplotlib.lines.Line2D at 0x7f8608a4f290>]
```



Calibration

The HBV model contains five stores where the water is stored and nine parameters that control the flow between those stores and in and out of the model. For the storages an array of starting values is specified. For the nine parameters, initial minimum and maximum values are stated. These values are based on ...

```
#           Si, Su, Sf, Ss, Sp
s_0 = np.array([0, 100, 0, 5, 0])
S_names = ["Interception storage", "Unsaturated Rootzone Storage", "Fastflow storage", "Slowfl

p_names = ["$I_{max}$", "$C_e$", "$Su_{max}$", " \beta", "$P_{max}$", "$T_{lag}$", "$K_f$
param_names = ["Imax","Ce", "Sumax", "beta", "Pmax", "Tlag", "Kf", "Ks", "FM"]\

# p_min_initial= np.array([0, 0.2, 40, .5, .001, 1, .01, .0001, .01])
# p_max_initial = np.array([8, 1, 800, 4, .3, 10, .1, .01, 0.5])

p_min_initial= np.array([0, 0.2, 40, .5, .001, 0.1, .01, .0001, .01])
p_max_initial = np.array([8, 4, 800, 4, 3, 10, 3, .1, 5])
```

The following part tries to find the best parameter combination using a randomizer. Later I used better ways so I commented this part because I don't think I will use it.

Randomizer To be able to find a good combination of parameters, a randomizer is used. It creates N possible combinations of parameters, with each of these parameters having a value in between the initial minimum and maximum values.

$N = 1000$

```
parameters = np.random.uniform(p_min_initial[:, np.newaxis], p_max_initial[:, np.newaxis], (le
# print(parameters)
```

The next step is to run the HBV model with the forcing, the randomly chosen parameters and the initially set storage data. First of all an empty list is made, called 'ensemble'. The HBV model is run N times, with every time another set of the randomly chosen parameters from above. The outcomes of the HBV model are stored in the ensemble list.

```
ensemble = []

for counter in range(N):
    ensemble.append(ewatercycle.models.HBVLocal(forcing=camelsgb_forcing))
    config_file, _ = ensemble[counter].setup(
        parameters = parameters[:,counter],
        initial_storage=s_0,
        cfg_dir = "configFiles/hbv_ensembleMember_" + str(counter),
    )
    ensemble[counter].initialize(config_file)
```

To be able to determine if the outcomes of the HBV model are any good, the results can be compared to the observations. It is important to first determine when the model works good. This research focusses on the flooding threshold of the U4 subway line. Flooding happens at high discharges, peaks. This implies that the model can be seen as good, when the amount and height of the modelles peaks are similar to the peaks in the observed data. This is done with the following function. The difference between the modelled data and the observed data is raised to the second power, to 'punish' large absolute differences more heavy. The absolute difference between the observed data and the modelled data is higher at the peaks. In this way the parameter combination is looked for which results in the 'best' model, defined above.

```
def calibrationObjective(modelOutput, observation, start_calibration, end_calibration):
    # a function that takes in two dataFrames, interpolates the model output to the
    # observations and calculates the average absolute difference between the two.

    #combine the two in one dataframe
    hydro_data = pd.concat([modelOutput.reindex(observation.index, method = 'ffill'), observat

    #only select the calibration period
    hydro_data = hydro_data[hydro_data.index > pd.to_datetime(pd.Timestamp(start_calibration)).
    hydro_data = hydro_data[hydro_data.index < pd.to_datetime(pd.Timestamp(end_calibration)).da

    #calculate mean absolute difference to the power 4, so peaks weigh more
    diff = (hydro_data['Q'] - hydro_data['model output']) ** 2
    absDiff = np.abs(diff)
    meanAbsDiff = np.mean(absDiff)

    return meanAbsDiff
```

Now the differences between the model and the observed data can actually be calculated. Every ensemble is compared to the observed data. The mean absolute differences between the modelled discharges and the observed discharges are called objectives.

```
#an object to show a progress bar, since this can take a while:
f = IntProgress(min=0, max=N) # instantiate the bar
display(f) # display the bar
```

```

#converting the observations to a pandas object
Q_pandas = ds_forcing["Q"].to_dataframe()
#an empty array to store the results in
objectives = []

#loop over all ensemble members
for ensembleMember in ensemble:
    Q_m = []
    time = []
    while ensembleMember.time < ensembleMember.end_time:
        ensembleMember.update()
        discharge_this_timestep = ensembleMember.get_value("Q")
        Q_m.append(discharge_this_timestep[0])
        time.append(pd.Timestamp(ensembleMember.time_as_datetime.date()))

    #calculate the objective function
    discharge_dataframe = pd.DataFrame({'model output': Q_m}, index=pd.to_datetime(time))
    objective_this_model = calibrationObjective(discharge_dataframe, Q_pandas["Q"], calibration_
    objectives.append(objective_this_model)

    #it is good practice to remove any variable you don't need anymore to save memory.
    del Q_m, time, discharge_dataframe, objective_this_model

    #update progress bar
    f.value += 1

# print(objectives)

IntProgress(value=0, max=1000)

#let's also show the minimal values:
parameters_minimum_index = np.argmin(np.array(objectives))

objective_minimum = np.min(objectives)

parameters_minimum = parameters[:,parameters_minimum_index]

print("Best parameter index:", parameters_minimum_index)
print("Best parameter set:", parameters_minimum)
print("Objective: ", objective_minimum)

Best parameter index: 403

Best parameter set: [1.30531138e -01 8.53231694e -01 1.56436715e+02 1.35969715e+00
 1.50986571e -02 1.04371241e+00 3.89692752e -01 6.22991134e -02
 8.83438042e -01]

Objective: 0.8989231889998252

#finally, just like before, we remove the models themselves to save up space and memory.
for ensembleMember in ensemble:
    ensembleMember.finalize()

```

Now the results can be analyzed. The parameter set that results in the lowest objective value, is the best found parameter combination to calibrate the model to the observations.

```

import matplotlib.pyplot as plt

# xFigNr = 2
# yFigNr = 4

# fig, axs = plt.subplots(xFigNr, yFigNr,figsize = (15,15))

# for xFig in range(xFigNr):
#     for yFig in range(yFigNr):
#         paramCounter = xFig*yFigNr + yFig
#         axs[xFig,yFig].plot(parameters[paramCounter,:],objectives,'.')
#         axs[xFig,yFig].set_title(p_names[paramCounter])

```

We can compare the modelled discharge to the observed discharge by plotting both. First a model object needs to be created. Following, a configuration file is created that contains information on the location of the forcing files and the values of the initial storages and the parameters. Then the model gets primed to start running through initializing.

```

par_0 = parameters_minimum
model = ewatercycle.models.HBVLocal(forcing=camelsgb_forcing)
config_file, _ = model.setup(parameters=par_0, initial_storage=s_0)
model.initialize(config_file)

```

Now the model can start running.

```

Q_m = []
time = []
while model.time < model.end_time:
    model.update()
    Q_m.append(model.get_value("Q")[0])
    time.append(pd.Timestamp(model.time_as_datetime))
model.finalize()

model_output = pd.Series(data=Q_m, name="Modelled_discharge", index=time)

```

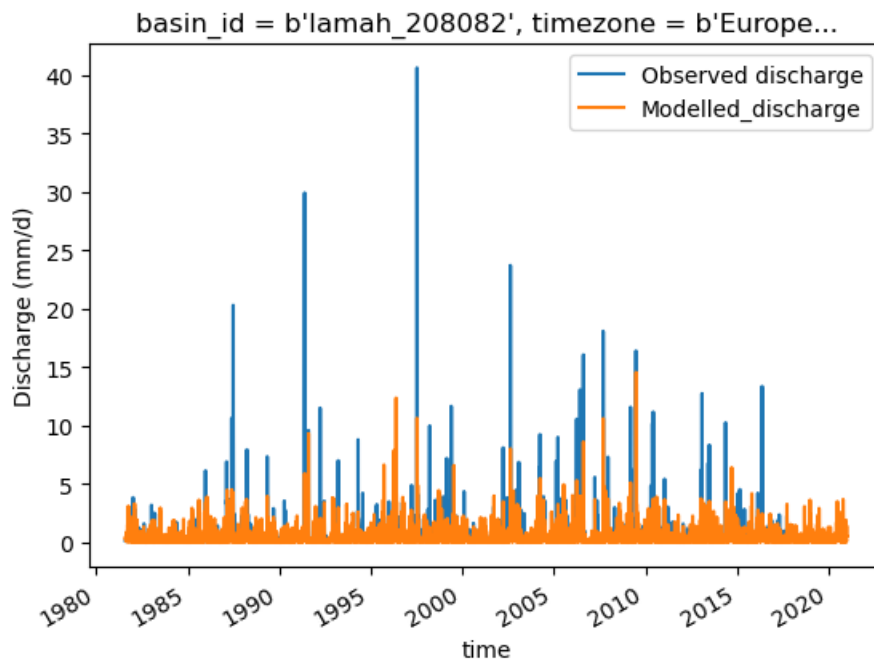
The modelled discharge and the observed discharge can be plotted to determine the accuracy of the model.

```

ds_forcing["Q"].plot(label="Observed discharge")
model_output.plot()
plt.legend()
plt.ylabel("Discharge (mm/d)")

Text(0, 0.5, 'Discharge (mm/d)')

```



To be able to use the output of the model in other analyses, it is saved as a NetCDF file:

```
xr_model_output = model_output.to_xarray()

xr_model_output.attrs['units'] = 'mm/d'

# Save the xarray Dataset to a NetCDF file
xr_model_output.to_netcdf('~river_discharge_data.nc')
```

Quantile Loss method and optimization with Nelder-Mead Another way of weighing the peaks more heavily is using the quantile loss method. The quantile loss method measures how well the model predicts a specific quantile by penalizing over- and under predictions more heavily. A quantile of more than 50 percent, weighs under predictions more heavily, and thus focusses more on the peaks.

Before, the parameter combinations were chosen randomly. A better way to predict the best parameter combination is through Nelder-Mead optimization. This optimization method finds the minimum of a function. The result of the Nelder-Mead optimization is the best parameter combination the method found. The Nelder-Mead optimization is run using the quantile loss method with quantiles of 90, 95 and 99 percent. All three the quantiles were used for now to see how they would result.

```
def calibrationObjective_quantile_loss(modelOutput, observation, start_calibration, end_calibr
    # a function that takes in two dataFrames, interpolates the model output to the
    # observations and calculates the average absolute difference between the two.

    #combine the two in one dataframe
    hydro_data = pd.concat([modelOutput.reindex(observation.index, method = 'ffill'), observat

    #only select the calibration period
    hydro_data = hydro_data[hydro_data.index > pd.to_datetime(pd.Timestamp(start_calibration).
    hydro_data = hydro_data[hydro_data.index < pd.to_datetime(pd.Timestamp(end_calibration).da

    #calculate mean absolute difference to the power 4, so peaks weigh more
    diff = (hydro_data['Q'] - hydro_data['model output'])
    absDiff = np.abs(diff)
    meanAbsDiff = np.mean(absDiff)
```

```

    return np.mean(np.maximum(q * diff, (q - 1) * diff)) # Quantile loss formule

#return meanAbsDiff

ensemble.append(ewatercycle.models.HBVLocal(forcing=camelsgb_forcing))
config_file, _ = ensemble[counter].setup(
    parameters = parameters[:,counter],
    initial_storage=s_0,
    cfg_dir = "configFiles/hbv_ensembleMember_" + str(counter),
)
ensemble[counter].initialize(config_file)

def objective_function(parameter_vector, q):
    model = ewatercycle.models.HBVLocal(forcing=camelsgb_forcing)
    config_file, _ = model.setup(
        parameters = parameter_vector,
        initial_storage=s_0,
        cfg_dir = "configFiles/hbv_ensembleMember_",
    )
    model.initialize(config_file)

    # discharge_this_timestep = model.get_value("Q")
    Q_m = []
    time = []
    while model.time < model.end_time:
        model.update()
        discharge_this_timestep = model.get_value("Q")
        Q_m.append(discharge_this_timestep[0])
        time.append(pd.Timestamp(model.time_as_datetime.date()))
    # model.finalize()
    # print(model.get_value("Q"))

    #calculate the objective function
    discharge_dataframe = pd.DataFrame({'model output': Q_m}, index=pd.to_datetime(time))

    objective_this_model = calibrationObjective_quantile_loss(discharge_dataframe, Q_pandas["Q"])
    del Q_m, time, discharge_dataframe

    if np.isnan(objective_this_model):
        return np.inf

    return objective_this_model

# objective_function(parameters_minimum)

p_min_initial= np.array([0, 0.2, 40, .5, .001, 1, .01, .0001, .01])
p_max_initial = np.array([50, 1, 800, 40, 3, 10, 1, .1, 0.9])

```

The next cell is commented because the cell takes very long to run, so I did not want to run it accidentally. I hardcoded the outcomes of the cell in the cells below.

```
# from scipy.optimize import minimize
```

```

# #bounds = [(0, 500), (0.2, 10),(40, 10000),(0.5, 10000),(0.001, 300),(1, 100),(0.01, 100),(0
# bounds = [(0, None), (0.001, None),(0.001, None),(0.001, None),(0.001, None),(0.001, None),(

# # result = minimize(objective_function, [2.10831791e+01, 2.85049614e -01, 2.05644822e+03, 2.
# # 2.90142667e+02, 1.37651503e+00, 2.39202301e+00, 1.26140676e+00,
# # 2.99032767e -01], bounds=bounds, method='Nelder -Mead', options={'xatol' : 0.000001})

# result90 = minimize(objective_function, [2.52268517e -05, 2.20175052e+00, 1.45074493e+02, 2.
# 1.01841470e+00, 6.84361550e -01, 5.85949449e -01, 4.42305303e -02,
# 1.05644244e+00], bounds=bounds, method='Nelder -Mead', options={'maxiter':3000}, args

# result95 = minimize(objective_function, [2.52268517e -05, 2.20175052e+00, 1.45074493e+02, 2.
# 1.01841470e+00, 6.84361550e -01, 5.85949449e -01, 4.42305303e -02,
# 1.05644244e+00], bounds=bounds, method='Nelder -Mead', options={'maxiter':3000}, args

# result99 = minimize(objective_function, [2.52268517e -05, 2.20175052e+00, 1.45074493e+02, 2.
# 1.01841470e+00, 6.84361550e -01, 5.85949449e -01, 4.42305303e -02,
# 1.05644244e+00], bounds=bounds, method='Nelder -Mead', options={'maxiter':3000}, args

result90

[5.58471586e -06,
 3.13412625,
 237.953048,
 5.73162134,
 0.652902354,
 1.45007782,
 0.504868854,
 0.0409678016,
 1.0886803]

result95

[1.11445722e -08,
 2.44479717,
 42.537041,
 1.2268688,
 2.09277774,
 1.04205225,
 0.727472601,
 0.0349031666,
 1.08389293]

result99

[1.04288193e -05,
 77.7284772,
 23.2176569,
 0.660999969,
 3.51075714,
 0.961766614,
 0.811764125,
 0.0370046705,
 1.00978819]

```



```

# print(result90.fun)
objective90 = 0.11857370061371166
# print(result95.fun)
objective95 = 0.08865588020728175
# print(result99.fun)
objective99 = 0.04227177059570846
# 0.9847863938751269  L -BFGS -B    kwadraat
#      0.78874967095675                Nelder -Mead    kwadraat

#1.4835419784065083  random met **4

# result90.x
# print(result90.x)
result90 = [5.58471586e -06, 3.13412625e+00, 2.37953048e+02, 5.73162134e+00,
6.52902354e -01, 1.45007782e+00, 5.04868854e -01, 4.09678016e -02,
1.08868030e+00]

# result95.x
# print(result95.x)
result95 = [1.11445722e -08, 2.44479717e+00, 4.25370410e+01, 1.22686880e+00,
2.09277774e+00, 1.04205225e+00, 7.27472601e -01, 3.49031666e -02,
1.08389293e+00]

# result99.x
# print(result99.x)
result99 = [1.04288193e -05, 7.77284772e+01, 2.32176569e+01, 6.60999969e -01,
3.51075714e+00, 9.61766614e -01, 8.11764125e -01, 3.70046705e -02,
1.00978819e+00]

par_90 = result90
model = ewatercycle.models.HBVLocal(forcing=camelsgb_forcing)
config_file, _ = model.setup(parameters=par_90, initial_storage=s_0)
model.initialize(config_file)

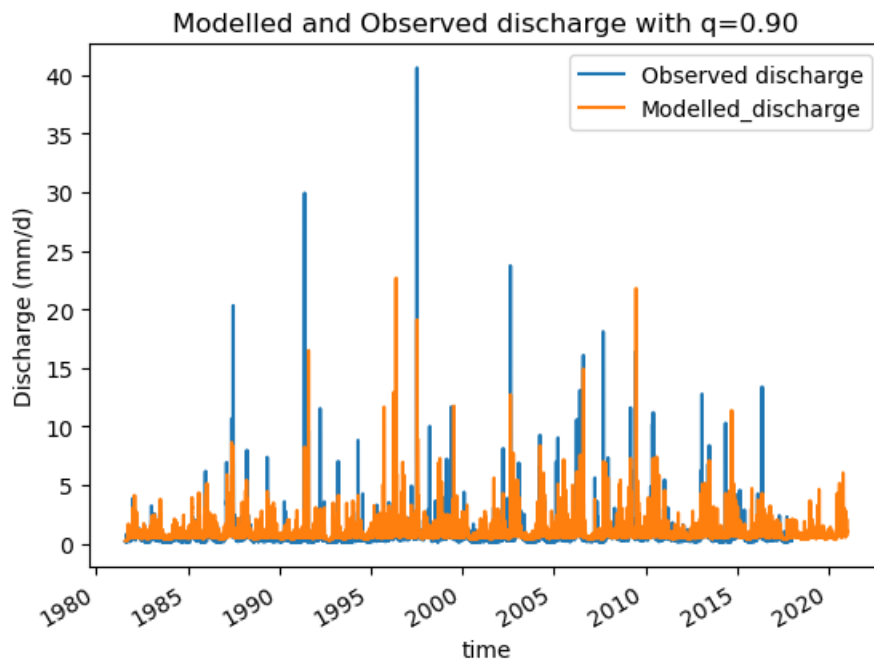
Q_m = []
time = []
while model.time < model.end_time:
    model.update()
    Q_m.append(model.get_value("Q")[0])
    time.append(pd.Timestamp(model.time_as_datetime))
model.finalize()

model_output90 = pd.Series(data=Q_m, name="Modelled_discharge", index=time)

ds_forcing["Q"].plot(label="Observed discharge")
model_output90.plot()
plt.legend()
plt.ylabel("Discharge (mm/d)")
plt.title('Modelled and Observed discharge with q=0.90')

Text(0.5, 1.0, 'Modelled and Observed discharge with q=0.90')

```



```

par_95 = result95
model = ewatercycle.models.HBVLocal(forcing=camelsgb_forcing)
config_file, _ = model.setup(parameters=par_95, initial_storage=s_0)
model.initialize(config_file)

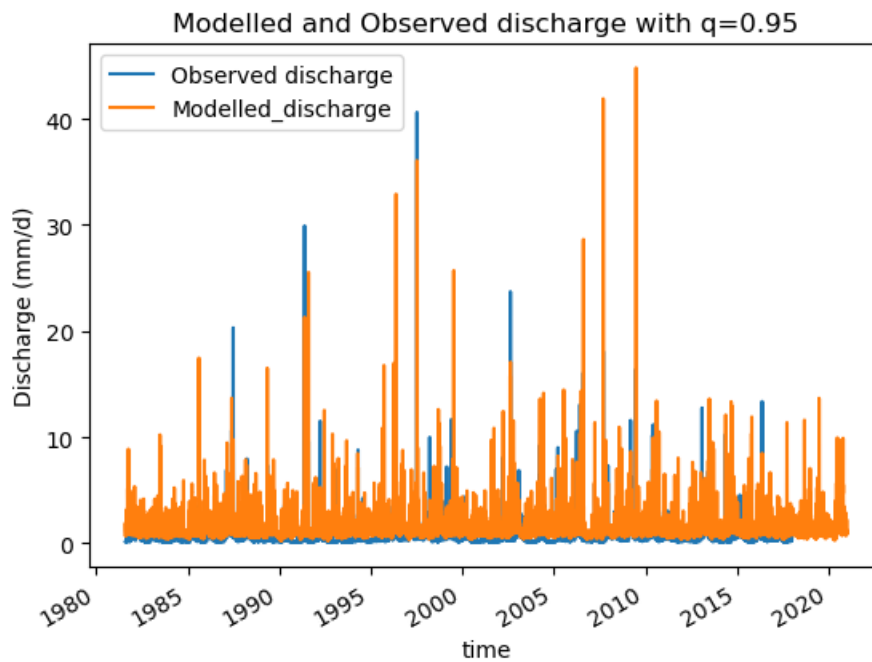
Q_m = []
time = []
while model.time < model.end_time:
    model.update()
    Q_m.append(model.get_value("Q")[0])
    time.append(pd.Timestamp(model.time_as_datetime))
model.finalize()

model_output95 = pd.Series(data=Q_m, name="Modelled_discharge", index=time)

ds_forcing["Q"].plot(label="Observed discharge")
model_output95.plot()
plt.legend()
plt.ylabel("Discharge (mm/d)")
plt.title('Modelled and Observed discharge with q=0.95')

Text(0.5, 1.0, 'Modelled and Observed discharge with q=0.95')

```



```

par_99 = result99
model = ewatercycle.models.HBVLocal(forcing=camelsgb_forcing)
config_file, _ = model.setup(parameters=par_99, initial_storage=s_0)
model.initialize(config_file)

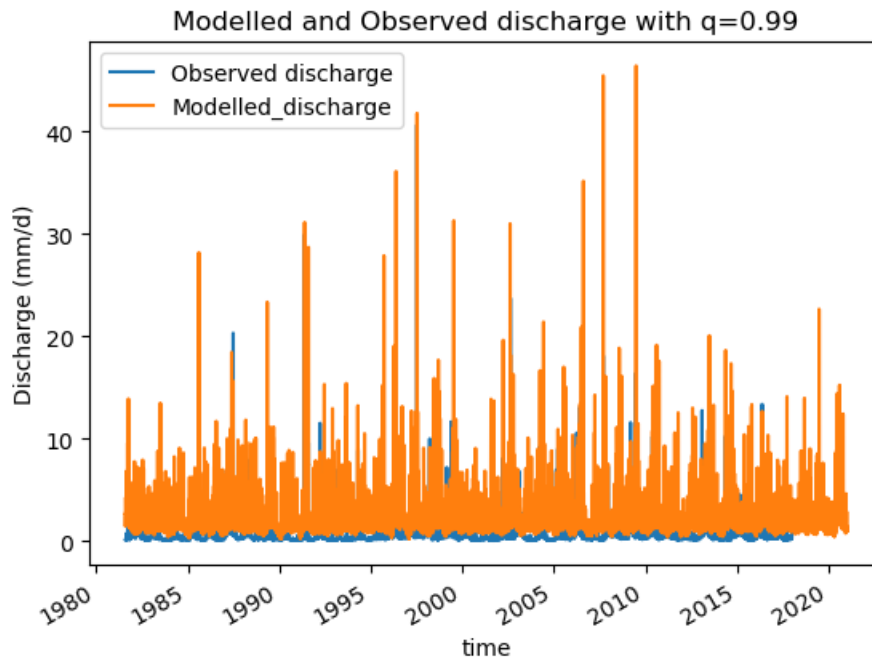
Q_m = []
time = []
while model.time < model.end_time:
    model.update()
    Q_m.append(model.get_value("Q")[0])
    time.append(pd.Timestamp(model.time_as_datetime))
model.finalize()

model_output99 = pd.Series(data=Q_m, name="Modelled_discharge", index=time)

ds_forcing["Q"].plot(label="Observed discharge")
model_output99.plot()
plt.legend()
plt.ylabel("Discharge (mm/d)")
plt.title('Modelled and Observed discharge with q=0.99')

Text(0.5, 1.0, 'Modelled and Observed discharge with q=0.99')

```



The next step is to validate the model output. This is done on the validation time period, which is the last 25 percent, so the last ten years, of the observed data. The mean absolute difference of the model output and the observed data, named the objective, of the calibration period can be compared to the objective of the validation period. The model is not overfitted for the calibration period, and thus usable for future predictions, if the objective values are similar.

```
def objective_function_validation(parameter_vector, q):
    model = ewatercycle.models.HBVLocal(forcing=camelsgb_forcing)
    config_file, _ = model.setup(
        parameters = parameter_vector,
        initial_storage=s_0,
        cfg_dir = "configFiles/hbv_ensembleMember_",
    )
    model.initialize(config_file)

    # discharge_this_timestep = model.get_value("Q")
    Q_m = []
    time = []
    while model.time < model.end_time:
        model.update()
        discharge_this_timestep = model.get_value("Q")
        Q_m.append(discharge_this_timestep[0])
        time.append(pd.Timestamp(model.time_as_datetime.date()))
    # model.finalize()
    # print(model.get_value("Q"))

    #calculate the objective function
    discharge_dataframe = pd.DataFrame({'model output': Q_m}, index=pd.to_datetime(time))

    objective_this_model = calibrationObjective_quantile_loss(discharge_dataframe, Q_pandas["Q"]
    del Q_m, time, discharge_dataframe

    if np.isnan(objective_this_model):
        return np.inf
```

```

    return objective_this_model

val90_objective = objective_function_validation([5.58471586e -06, 3.13412625e+00, 2.37953048e+
6.52902354e -01, 1.45007782e+00, 5.04868854e -01, 4.09678016e -02,
1.08868030e+00], 0.90)
print(objective90)
print(val90_objective)

0.11857370061371166

0.09339490016948018

val95_objective = objective_function_validation([1.11445722e -08, 2.44479717e+00, 4.25370410e+
2.09277774e+00, 1.04205225e+00, 7.27472601e -01, 3.49031666e -02,
1.08389293e+00], 0.95)
print(objective90)
print(val95_objective)

0.11857370061371166

0.07222547672725184

val99_objective = objective_function_validation([1.04288193e -05, 7.77284772e+01, 2.32176569e+
3.51075714e+00, 9.61766614e -01, 8.11764125e -01, 3.70046705e -02,
1.00978819e+00], 0.99)
print(objective99)
print(val99_objective)

0.04227177059570846

0.0339318687843097

```

It can be seen that for all three models, the objective for the validation period is lower than the objective for the training period. This would mean that the model is not overfit on the trainingsdata, since it is able to make good predictions on unseen data. The fact that the objective is lower for the testdata could mean that the testdata contains data that is easier to predict. It can be seen in the graphs above, that the observed data in the test period, so from 2010 to 2021, does not contain peaks as high as in the training period, from 1980 to 2010. The model might be improved by using cross validation.

Kling Gupta efficiency For my research, the height and frequency of the peaks are important, but their timing is less critical. The Kling-Gupta Efficiency is a measure that evaluates how well a model performs by looking at correlation, bias, and variability. It is especially useful for predicting the size of peaks and the overall distribution, while being less focused on the exact timing of the peaks.

```

import numpy as np

def kling_gupta_efficiency(obs, sim, start_calibration, end_calibration):

    #combine the two in one dataframe
    hydro_data = pd.concat([sim.reindex(obs.index, method = 'ffill'), obs], axis=1)

    #only select the calibration period
    hydro_data = hydro_data[hydro_data.index > pd.to_datetime(pd.Timestamp(start_calibration)).
    hydro_data = hydro_data[hydro_data.index < pd.to_datetime(pd.Timestamp(end_calibration)).da

```

```

# #calculate mean absolute difference to the power 4, so peaks weigh more
# diff = (hydro_data['Q'] - hydro_data['model output'])
# absDiff = np.abs(diff)
# meanAbsDiff = np.mean(absDiff)

r = np.corrcoef(hydro_data['Q'], hydro_data['model output'])[0, 1] # Correlation
beta = np.mean(hydro_data['model output']) / np.mean(hydro_data['Q']) # Bias ratio
gamma = np.std(hydro_data['model output']) / np.std(hydro_data['Q']) # Variability ratio

return -(1 - np.sqrt((r - 1) ** 2 + (beta - 1) ** 2 + (gamma - 1) ** 2))

# objective = kling_gupta_efficiency(observations, model_output)

def objective_function(parameter_vector):
    model = ewatercycle.models.HBVLocal(forcing=camelsgb_forcing)
    config_file, _ = model.setup(
        parameters = parameter_vector,
        initial_storage=s_0,
        cfg_dir = "configFiles/hbv_ensembleMember_",
    )
    model.initialize(config_file)

    # discharge_this_timestep = model.get_value("Q")
    Q_m = []
    time = []
    while model.time < model.end_time:
        model.update()
        discharge_this_timestep = model.get_value("Q")
        Q_m.append(discharge_this_timestep[0])
        time.append(pd.Timestamp(model.time_as_datetime.date()))
    # model.finalize()
    # print(model.get_value("Q"))

    #calculate the objective function
    discharge_dataframe = pd.DataFrame({'model output': Q_m}, index=pd.to_datetime(time))

    objective_this_model = kling_gupta_efficiency(discharge_dataframe, Q_pandas["Q"], calibrat
del Q_m, time, discharge_dataframe

    if np.isnan(objective_this_model):
        return np.inf

    return objective_this_model

from scipy.optimize import minimize

#bounds = [(0, 500), (0.2, 10),(40, 10000),(0.5, 10000),(0.001, 300),(1, 100),(0.01, 100),(0.0
bounds = [(0, None), (0.001, None),(0.001, None),(0.001, None),(0.001, None),(0.001, None),(0.

result90 = minimize(objective_function, [2.52268517e -05, 2.20175052e+00, 1.45074493e+02, 2.48
1.01841470e+00, 6.84361550e -01, 5.85949449e -01, 4.42305303e -02,

```

```

1.05644244e+00], bounds=bounds, method='Nelder -Mead', options={'maxiter':100})

result90

message: Maximum number of iterations has been exceeded.
      success: False
      status: 2
      fun: -0.5036647074766698
      x: [ 3.257e -05  1.556e+00  1.405e+02  3.016e+00  2.671e -01
           6.978e -01  8.097e -01  3.995e -02  1.141e+00]
      nit: 100
      nfev: 144
final_simplex: (array([[ 3.257e -05,  1.556e+00, ...,  3.995e -02,
                        1.141e+00],
                       [ 3.200e -05,  1.528e+00, ...,  4.211e -02,
                        1.016e+00],
                       ...,
                       [ 3.168e -05,  1.594e+00, ...,  4.188e -02,
                        1.015e+00],
                       [ 3.177e -05,  1.529e+00, ...,  4.242e -02,
                        1.013e+00]]), array([-5.037e -01, -5.014e -01, -5.007e -01, -5.006e
-4.994e -01, -4.993e -01, -4.993e -01, -4.983e -01,
-4.982e -01, -4.979e -01]))

result90.x

array([3.25683830e -05, 1.55625787e+00, 1.40454772e+02, 3.01564724e+00,
       2.67053370e -01, 6.97751529e -01, 8.09680964e -01, 3.99505296e -02,
       1.14067403e+00])

result90.fun

-0.5036647074766698

par_99 = result90.x
model = ewatercycle.models.HBVLocal(forcing=camelsgb_forcing)
config_file, _ = model.setup(parameters=par_99, initial_storage=s_0)
model.initialize(config_file)

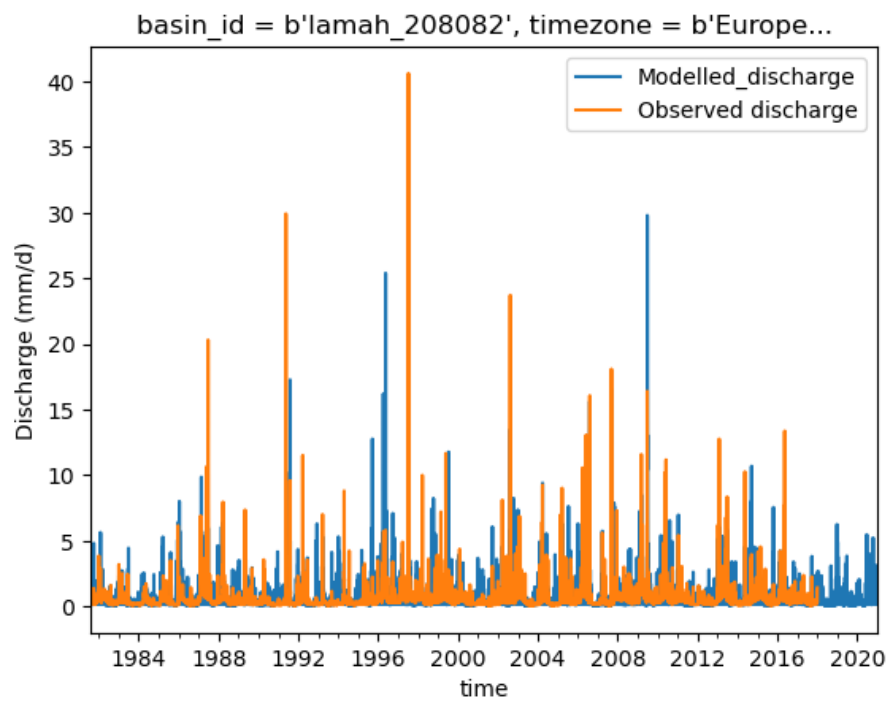
Q_m = []
time = []
while model.time < model.end_time:
    model.update()
    Q_m.append(model.get_value("Q")[0])
    time.append(pd.Timestamp(model.time_as_datetime))
model.finalize()

model_output99 = pd.Series(data=Q_m, name="Modelled_discharge", index=time)

model_output99.plot()
ds_forcing["Q"].plot(label="Observed discharge")

plt.legend()
plt.ylabel("Discharge (mm/d)")

Text(0, 0.5, 'Discharge (mm/d)')
```



to consider:

- cross validation
- which one to use eventually

0.1.5 Chapter 4: Future frequency of threshold exceedance under different climate change scenarios

0.1.6 References

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